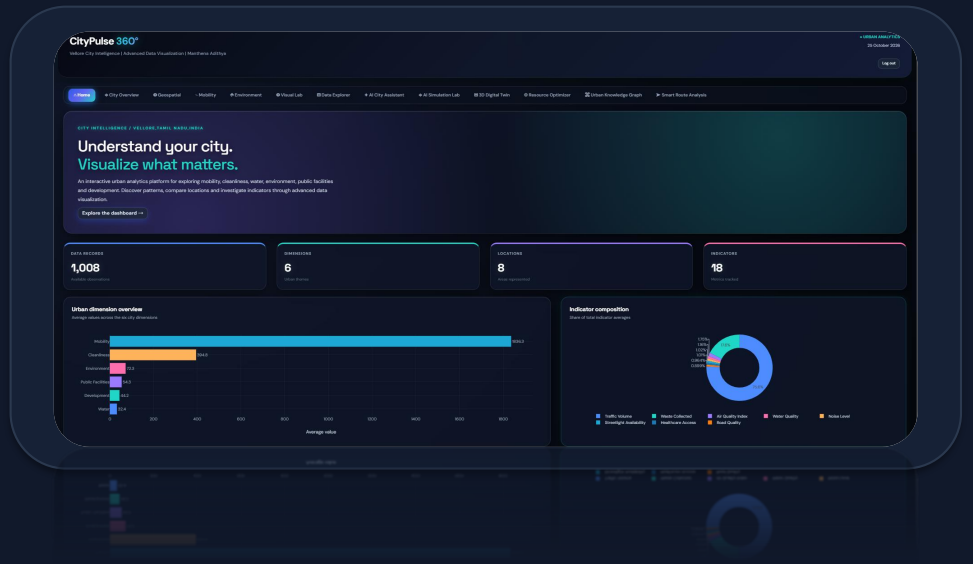
CSI3005 • FALL SEMESTER 2026–2027

CityPulse 360°

Urban visual analytics for mobility and city-service decisions



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All example values are synthetic demonstration data unless independently verified.

Assessment scope and structure

This presentation follows the four tasks and mark allocation in the provided assessment brief.

ASSESSMENT TASKS

- Task 1 — Frame the problem (2 marks)
- Task 2 — Choose and defend two visualizations (3 marks)
- Task 3 — Critique and correct a flawed design (3 marks)
- Task 4 — Adapt one visualization for streaming data (2 marks)

DELIVERABLES

- Case study context, audience and decision objective
- Dataset source and relevant attributes
- Two analytical questions
- Labelled sketches, comparisons and design justification
- Flawed and corrected designs plus streaming proposal

Case study context

CityPulse 360° is a prototype for comparing urban indicators across monitored Vellore locations.

- Urban managers need to compare mobility, cleanliness, water, environment, public facilities and development indicators.
- Different services generate different kinds of data and use different measurement units.
- A dashboard should help locate high-priority areas, not merely display many charts.
- This case study focuses on spatial comparison and changes over time.

Intended audience

The audience determines which comparisons are useful and how much detail is appropriate.

PRIMARY USERS

- Municipal planners and decision-makers
- Traffic and transport engineers
- Sanitation, water and environmental teams
- Analysts preparing reports for city administrators

USER NEEDS

- Quickly identify high-need locations
- Compare indicators without confusing units
- Inspect unusual patterns and data limitations
- Prioritize field visits and resource allocation

Decision objective

The visualizations support investigation and prioritization; they do not automatically prove causes.

- Compare locations to identify where traffic pressure appears highest.
- Screen environmental and water indicators to find unusual locations requiring inspection.
- Use the results to plan targeted data collection or operational follow-up.
- Validate high-impact findings with verified sensor records and local expertise before acting.

Dataset source and limitations

Source: CityPulse 360° prototype screens supplied for this assessment.

- Dataset source: the case-study prototype and its displayed demonstration dataset.
- The interface reports 1,008 records, 6 dimensions, 8 locations and 18 indicators.
- The prototype explicitly labels its values as synthetic demonstration data.
- Therefore, results in this presentation demonstrate visualization reasoning; they are not verified live city facts.
- For operational deployment, use documented municipal sources, sensor metadata, collection dates and quality checks.

Relevant dataset attributes

Identify the data type of each field before choosing visual encodings.

ATTRIBUTE & TYPE

- Location — nominal category
- Date — temporal
- Urban dimension — nominal category
- Indicator — nominal category
- Value — quantitative numeric measure
- Source / quality flag — provenance and reliability

VISUALIZATION ROLE

- Location defines groups or map positions
- Date supports trends and streaming windows
- Dimension and indicator support filtering
- Value controls bar length, line height or colour intensity
- Source/quality flags help distinguish measured and synthetic data

Analytical question 1

Question 1: Where is traffic pressure highest across monitored locations?

- Question: Which locations show the highest traffic-volume indicator, and how do they compare with the other locations?
- Analytical task: rank and compare quantitative values across discrete categories.
- Primary visual: sorted horizontal bar chart with a zero baseline.
- Decision relevance: identify candidate areas for traffic counts, congestion audits or follow-up monitoring.
- Caution: confirm what the traffic-volume metric measures before treating it as vehicle count.

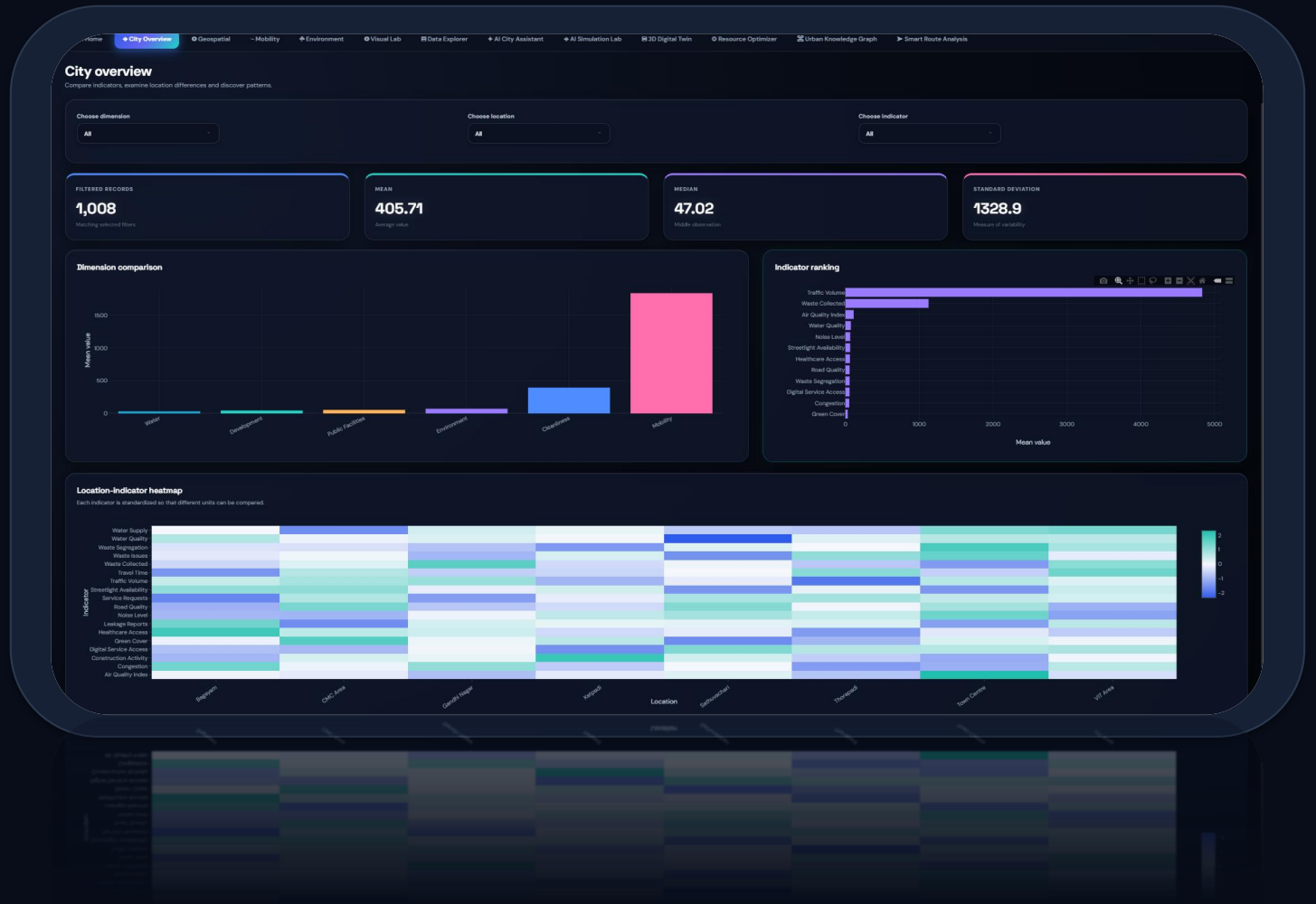
Analytical question 2

Question 2: How do environmental and water indicators vary by location?

- Question: Which locations are above or below the city-wide pattern across multiple environmental and water indicators?
- Analytical task: find multivariate patterns and outliers across categories.
- Primary visual: standardized location-by-indicator heatmap.
- Decision relevance: focus inspection on locations with unusual values across relevant services.
- Caution: standardize within each indicator because raw units may not be comparable.

Visualization 1 — ranked horizontal bars

Proposed chart for Question 1: compare traffic volume across locations.



ANALYSIS & JUSTIFICATION

- Y-axis: location names, sorted from highest to lowest value.
- X-axis: traffic-volume measure, beginning at zero; include a unit or explicit index label.
- Encoding: bar length and position encode quantitative magnitude.
- Direct labels may show values; use one consistent bar colour and a restrained highlight for the highest value.
- The sorted order reduces search effort and makes ranking immediate.

Values shown in the prototype are synthetic; confirm metric definitions.

Why bars suit Question 1

Bar length uses a common baseline, supporting accurate comparisons.

ADVANTAGES

- Works well for one quantitative measure across nominal locations
- Sorting reveals rank order immediately
- Labels can preserve exact comparisons
- Horizontal layout accommodates longer place names

DESIGN SAFEGUARDS

- Start the value axis at zero
- State the measure and units
- Avoid 3D bars and decorative gradients
- Use consistent scale across comparisons

Alternative to bars: pie chart

A pie chart is less suitable for comparing close location values.

PIE CHART LIMITATIONS

- Angles and areas are harder to compare precisely
- Eight similar slices create visual clutter
- Rank order is not immediately visible
- A pie suggests parts of one whole, which may not be the analytical goal

WHY BARS WIN

- Bar lengths share a baseline
- Small differences remain easier to compare
- Descending order supports prioritization
- The chart answers ranking rather than share-of-whole questions

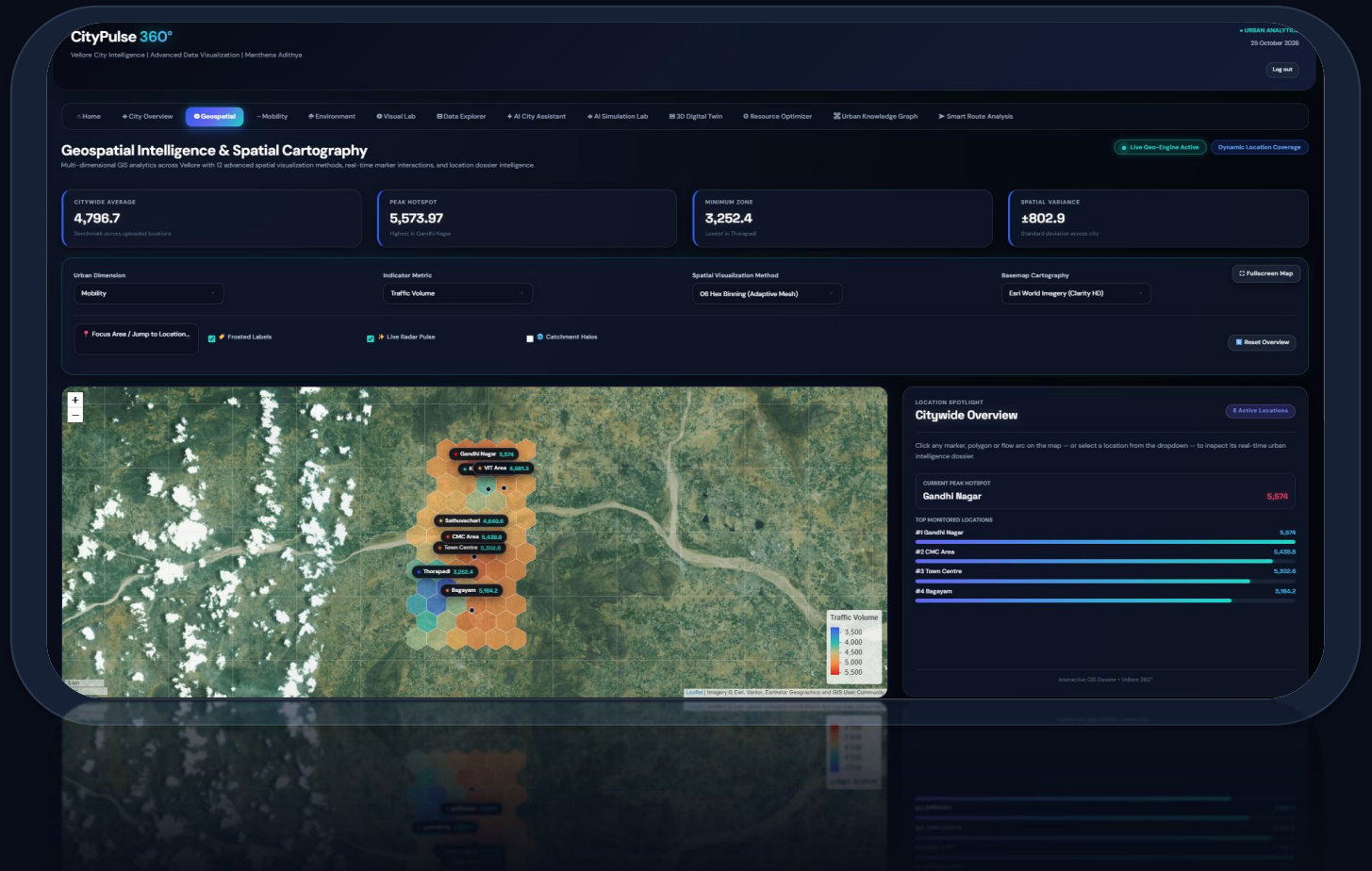
Bar-chart sketch and labels

Sketch specification for the report: draw and label axes, values and ordering.

- Title: Mean traffic-volume indicator by location.
- Horizontal axis: Traffic-volume indicator (synthetic index; zero baseline).
- Vertical axis: location, ordered from largest to smallest.
- Use a consistent scale and value labels at bar ends.
- Add a note: “Illustrative values from synthetic demonstration data.”

Visualization 2 — standardized heatmap

Proposed chart for Question 2: compare patterns across multiple indicators and locations.



ANALYSIS & JUSTIFICATION

- Columns: eight locations; rows: selected environment and water indicators.
- Cell colour represents a standardized value, such as a z-score within each indicator.
- Use a diverging colour scale centred at zero: below average, near average, above average.
- Include a legend and explain the standardization method.
- Cluster or group related indicators only if the ordering improves interpretation.

Heatmap colours must represent a defined scale, not decoration.

Why a heatmap suits Question 2

A heatmap supports rapid pattern detection across many rows and columns.

STRENGTHS

- Compact view of multiple indicators
- Makes unusual location-indicator combinations visible
- Supports screening before detailed investigation
- Works with categorical rows/columns and quantitative colour encoding

LIMITATIONS & CONTROLS

- Exact values are less precise than labels or bars
- Colour perception varies; use a colour-blind-aware palette
- Normalize indicators with different units
- Provide tooltips or a companion table for exact values

Alternative to heatmap: small multiples

Small-multiple bar charts are useful for exact comparisons but take more space.

SMALL MULTIPLES

- One chart per indicator
- Clear axis and units for each measure
- Good when precise values are the priority
- Can require extensive scrolling or page space

HEATMAP ADVANTAGE

- One compact overview
- Makes cross-location patterns easier to scan
- Best for exploratory screening
- Detailed follow-up can use small multiples or tables

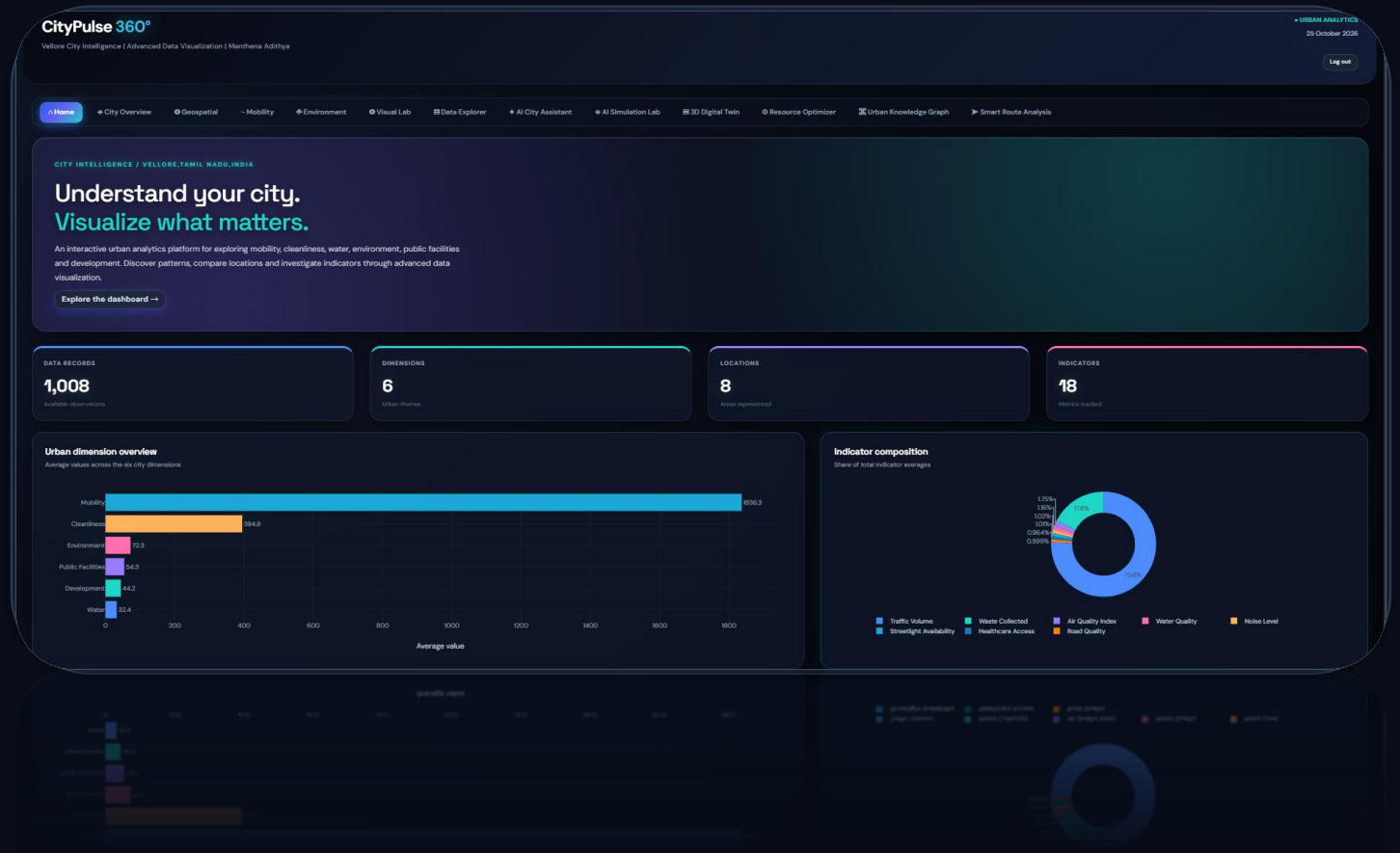
Heatmap sketch specification

A correct sketch must make the mapping and scale explicit.

- Title: Standardized environmental and water indicators by location.
- X-axis: location names.
- Y-axis: indicator names.
- Cell fill: standardized value; diverging palette centred on zero.
- Legend: below average → average → above average.
- Note: standardized comparisons indicate relative position, not absolute severity.

Prototype evidence — city overview

The homepage summarizes the number of records, dimensions, locations and indicators.

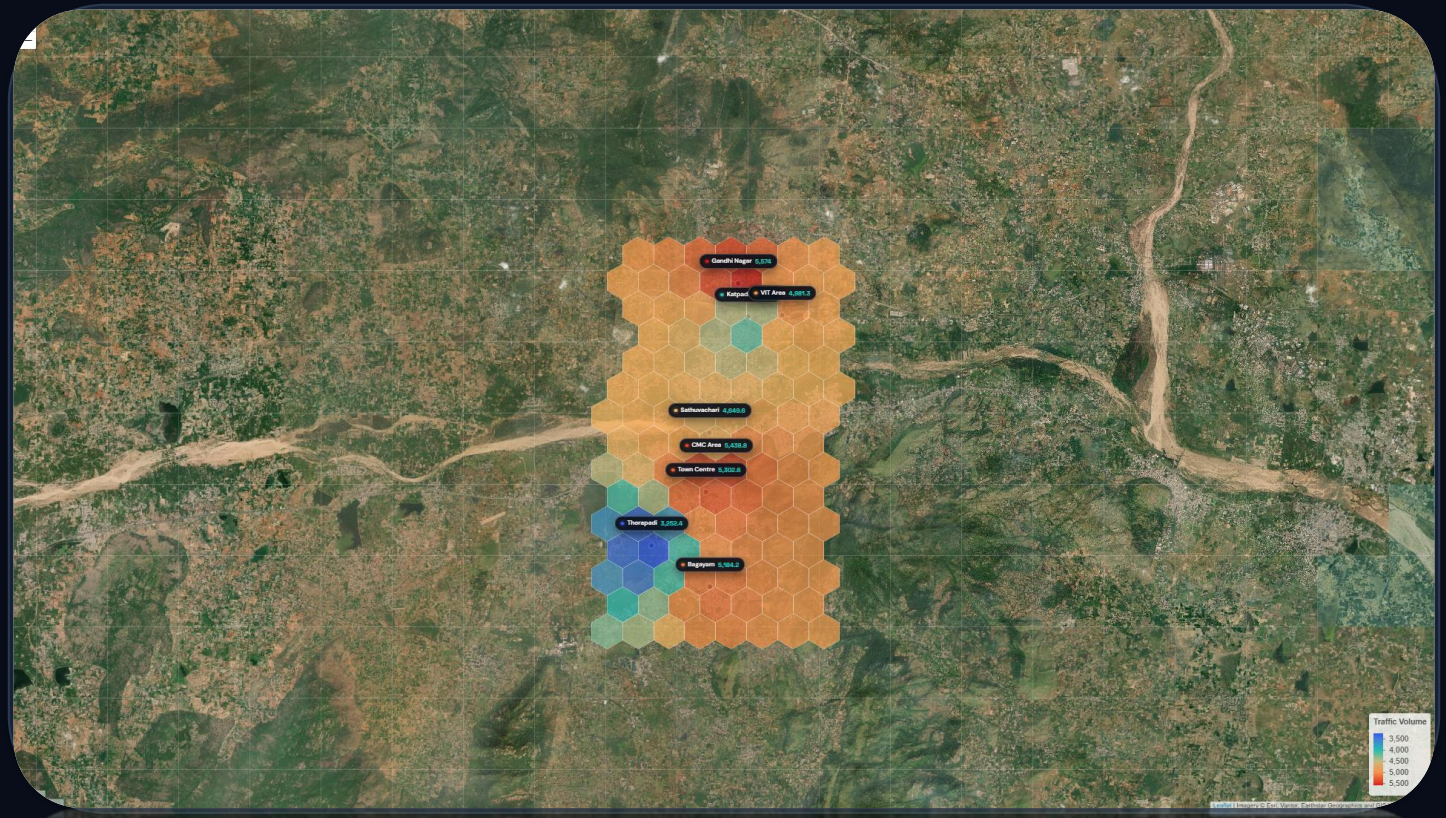


ANALYSIS & JUSTIFICATION

- The summary cards provide a quick overview of dataset coverage.
- The urban-dimension bar chart supports broad comparison across dimensions.
- The indicator-composition donut summarizes the distribution of selected indicator averages.
- Potential improvement: label units and explain why aggregation across different indicators is meaningful.
- Keep the home page as an overview; detailed questions belong in focused analytical views.

Prototype evidence — geospatial view

The geospatial screen adds spatial context to the location-level indicator values.



ANALYSIS & JUSTIFICATION

- Map encodings can reveal where values cluster geographically.
- A map is useful when physical proximity or catchment area affects decisions.
- A legend should identify the metric and direction of high/low values.
- Map labels must not overlap excessively; offer selection or zoom for dense regions.
- The shown coordinates and indicator values are approximate/synthetic and require validation.

Prototype evidence — mobility and environment

Time-series and distribution views add temporal and variability context.



ANALYSIS & JUSTIFICATION

- A line chart is appropriate when the horizontal axis is time and order matters.
- A box plot can summarize spread, median and potential outliers.
- Avoid combining measures with vastly different units on one unscaled axis.
- Show date range, sampling frequency and missing-data treatment.
- Use these views for deeper investigation after the primary comparison.

Critique task — deliberately flawed chart

Flaw 1: truncated quantitative axis exaggerates modest differences.

- Construct a bar chart whose axis starts near the smallest value instead of zero.
- Use traffic values that are close to one another to make the distortion visible.
- The tallest bar appears disproportionately larger than the others.
- The visual exaggeration can lead decision-makers to overstate the gap between locations.
- Clearly mark this chart as intentionally flawed in the assessment.

Flaw 2 — ambiguous units and metric

A chart can mislead even when the axis is not truncated.

- Use an unclear title such as “Traffic” without defining the measure.
- Omit the unit or whether the value is a count, rate or synthetic index.
- Viewers may interpret a modelled index as a verified vehicle count.
- This weakens comparability and can lead to incorrect public claims.
- The flaw is semantic: the chart does not explain what the numbers mean.

Flawed chart — combined critique

The flawed design uses two ineffective choices at once.

FLAW

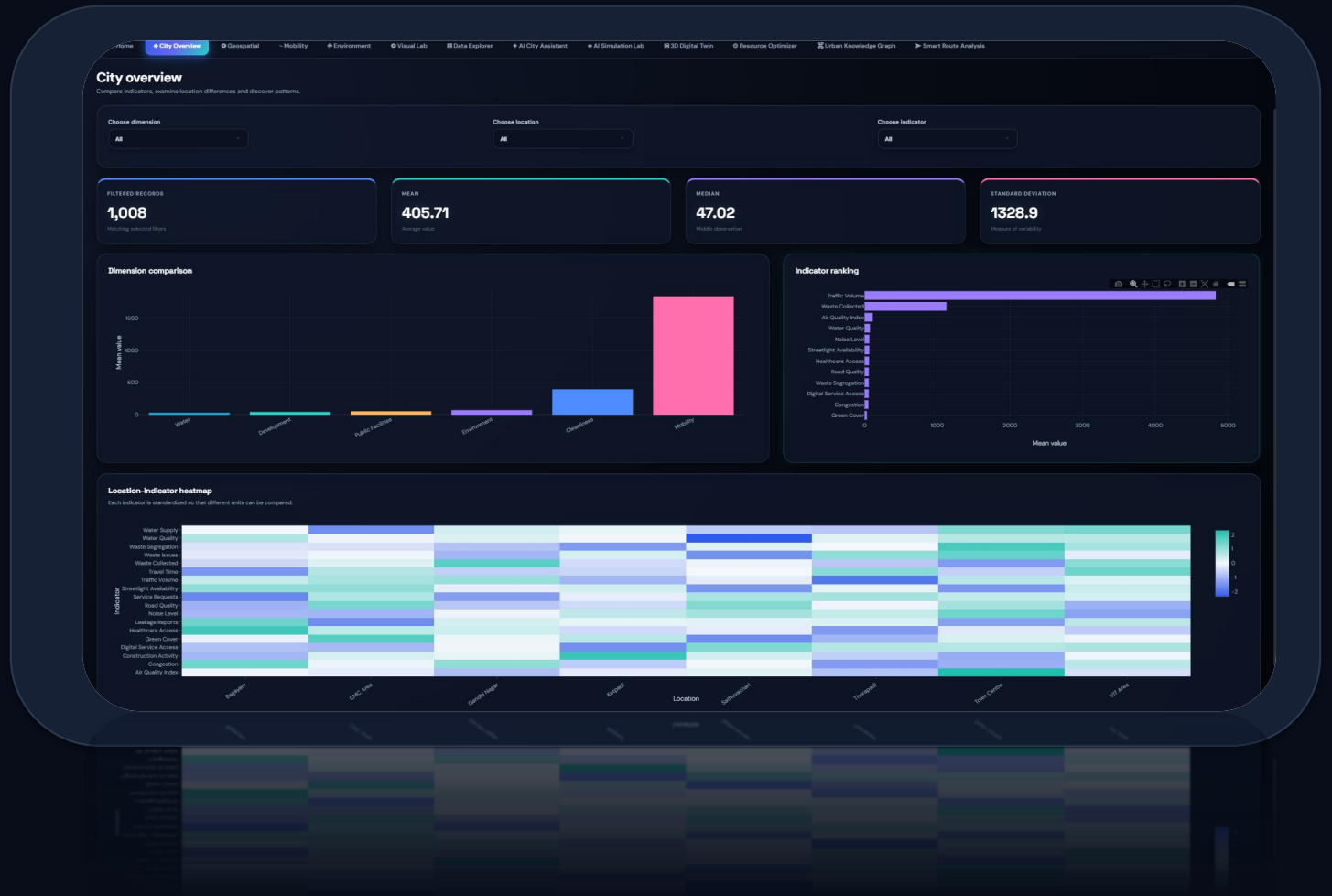
- Truncated value axis
- Ambiguous indicator name or missing unit
- No source or data-status note

LIKELY EFFECT

- Overstates small differences
- Encourages false precision
- Makes synthetic data look like verified observations
- Can distort resource-prioritization decisions

Corrected chart — zero baseline

Restore a truthful scale and make the metric interpretable.



ANALYSIS & JUSTIFICATION

- Start the quantitative axis at zero for a bar chart.
- Use a specific title such as “Traffic-volume indicator by location.”
- Label the axis with the measure and unit, or explicitly say “synthetic index.”
- Order locations consistently and show values where useful.
- Include the source, date range and data limitation note.

A zero baseline is appropriate for bar-length comparisons.

Why the correction improves accuracy

The corrected design reduces distortion and improves the basis for decisions.

BEFORE

- Visual differences exaggerated by truncated scale
- Meaning of values unclear
- Source and data status hidden

AFTER

- Zero baseline supports proportional comparison
- Metric and unit are explicit
- Synthetic status and source are stated
- Readers can judge evidence limits before acting

Changed requirement — continuous monitoring

The stakeholder now needs to monitor traffic as new data arrive.

- Adapt the traffic-volume comparison into a time-series line chart for a selected location or a small set of locations.
- Use time on the horizontal axis and traffic volume on the vertical axis.
- Preserve a fixed time scale so changes can be compared over time.
- Show a threshold only if it is based on a validated operational standard or baseline.
- Allow users to pause, inspect a point and compare with a daily summary.

Update interval and display window

Recommended starting configuration: one-minute updates and a rolling 60-minute window.

PROPOSED SETTINGS

- Refresh: every 1 minute, subject to source latency
- Display: most recent 60 minutes
- Keep longer daily/weekly summaries separately
- Annotate missing or delayed readings

RATIONALE

- Frequent updates help detect sudden spikes
- A one-hour window reveals short-term trend and context
- A separate daily view avoids losing long-term patterns
- Do not refresh faster than the source can reliably deliver

Raw values versus aggregation

Combine recent individual observations with a smoothed trend.

INDIVIDUAL VALUES

- Expose sudden changes and outliers
- Useful for incident detection
- Can look noisy when measurements fluctuate rapidly

AGGREGATED VALUES

- A 5-minute rolling average reveals underlying trend
- Reduces short-term noise
- May delay or hide brief spikes
- Show raw line/points alongside the smoothed line

Trade-off and operational safeguards

The display should balance rapid detection with trend readability.

- A shorter update interval reduces detection delay but increases visual churn and system load.
- A longer time window gives context but compresses small events.
- Smoothing improves readability but can hide brief peaks; retain raw observations.
- Use alert thresholds only after validating baseline behaviour and operational consequences.
- Display timestamp, data freshness and outage status to avoid false reassurance.

Conclusion and assessment checklist

CityPulse 360° links visualization choices to specific urban decisions.

MAIN CONCLUSIONS

- Sorted horizontal bars answer location-ranking questions
- Standardized heatmaps reveal patterns across unlike indicators
- Honest scales, labels and source notes reduce misleading interpretation
- Streaming line charts balance rapid detection and trend context

BRIEF REQUIREMENTS COVERED

- Task 1: context, audience, attributes and two questions
- Task 2: two techniques, labelled design logic and alternatives
- Task 3: two flaws, impacts and corrected design
- Task 4: update interval, time window and aggregation trade-off